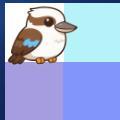




Centre for
Robotics



SLAM Course

05 - Overview

Dr Alejandro Fontán Villacampa
April 2026



Dr. Alejandro Fontán

**Bachelor and Master in
Mechanical Engineering**



**Universidad
Zaragoza**

**PhD -
Information Driven Navigation**



**Deutsches Zentrum
für Luft- und Raumfahrt**
German Aerospace Center



Postdoc in QCR



Internship at AR company

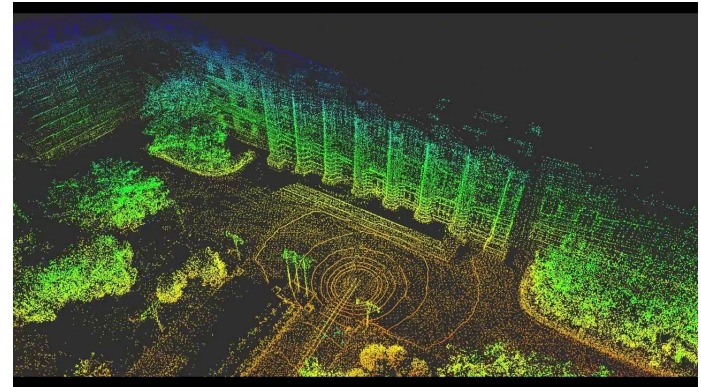


wikitude



Lecture Outline - Simultaneous Localization & Mapping

- ❖ What is SLAM?
 - Mapping
 - Localization
- ❖ Anatomy of a Modern SLAM System
 - Sensors
 - Front-End (Indirect / Direct)
 - Odometry Estimation
 - Loop Closure
 - Back-End
 - Geometry Consistency
 - State Estimation Techniques
- ❖ Why do we need SLAM?
- ❖ Do we really need SLAM for Robotics?
- ❖ SLAM Applications
- ❖ SLAM Systems



What is SLAM? - Mapping

Mapping is the process of building a spatial representation of an unknown or partially known environment.

- **Metric**: Precise geometric information (e.g., x,y coordinates)
- **Topological**: Relational information (e.g., a subway map)

Representations:

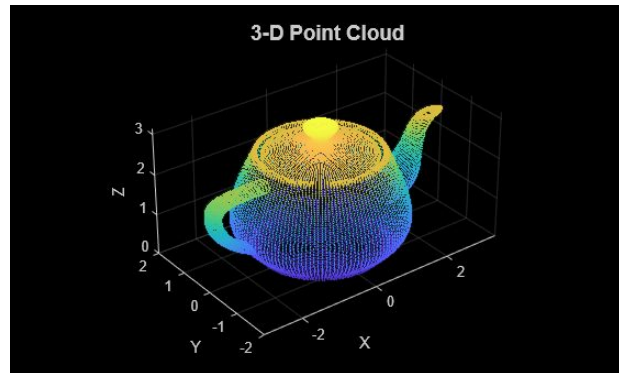
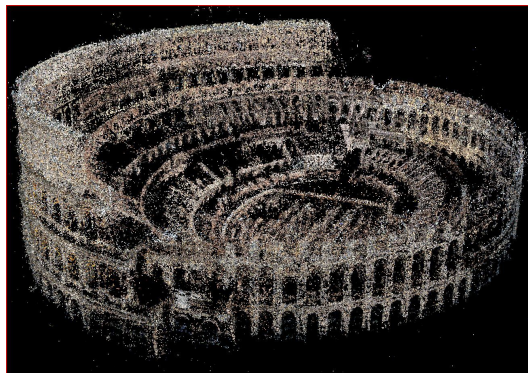
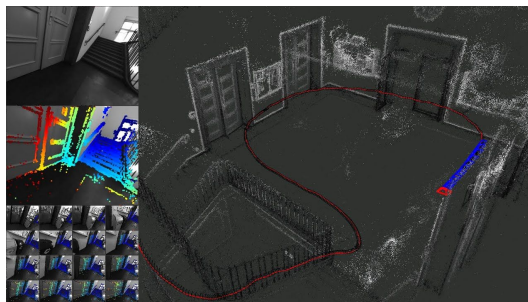
- Point Clouds
- Polygons / patches
- Occupancy grid / Voxels
- Semantics / objects
- Neural Radiance Fields (NeRFs)
- Gaussian Splatting
- Foundation Models
- ...



What is SLAM? - Mapping

Representations:

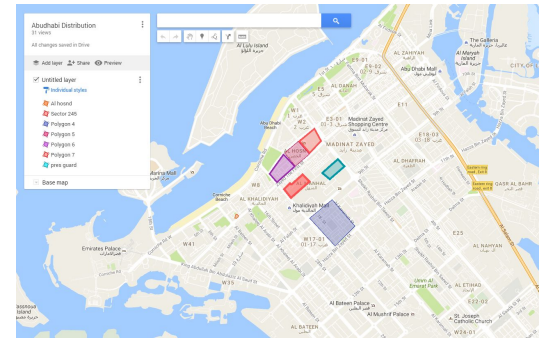
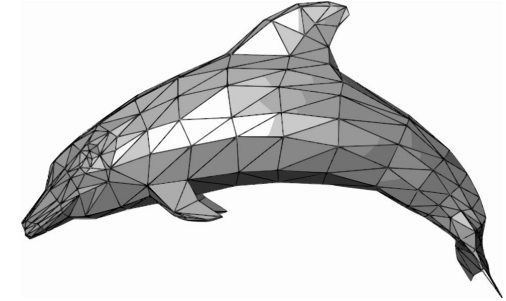
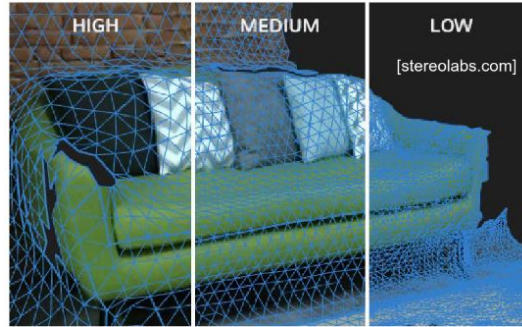
- ❖ **Point Clouds**
- ❖ Polygons / patches
- ❖ Occupancy grid
- ❖ Semantics / objects
- ❖ NeRFs
- ❖ Gaussian Splatting
- ❖ Foundation Models
- ❖ ...



What is SLAM? - Mapping

Representations:

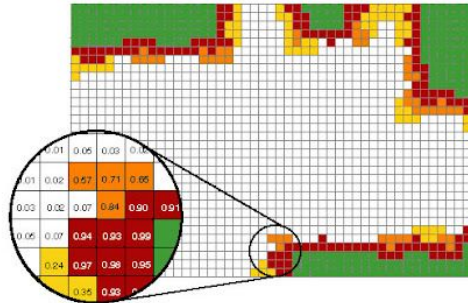
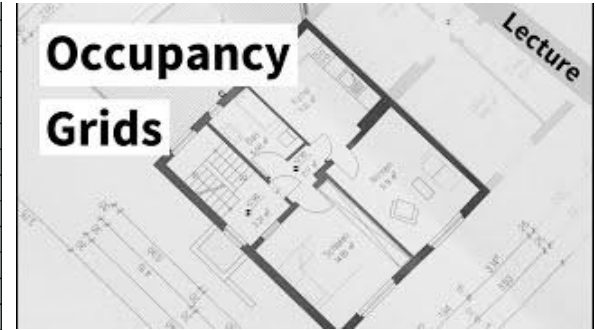
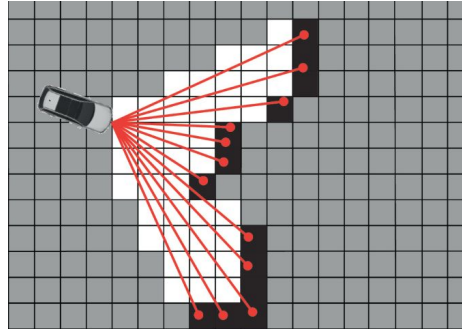
- ❖ Point Clouds
- ❖ **Polygons / patches**
- ❖ Occupancy grid / Voxels
- ❖ Semantics / objects
- ❖ NeRFs
- ❖ Gaussian Splatting
- ❖ Foundation Models
- ❖ ...



What is SLAM? - Mapping

Representations:

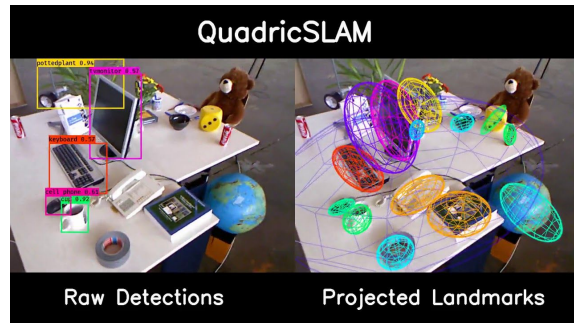
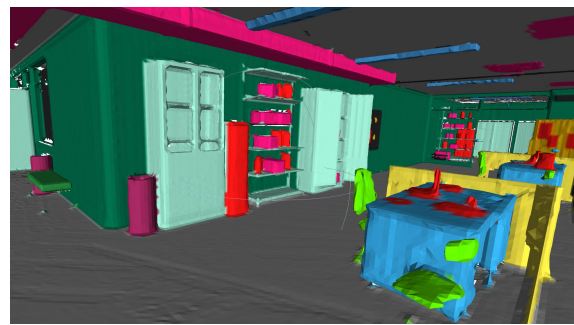
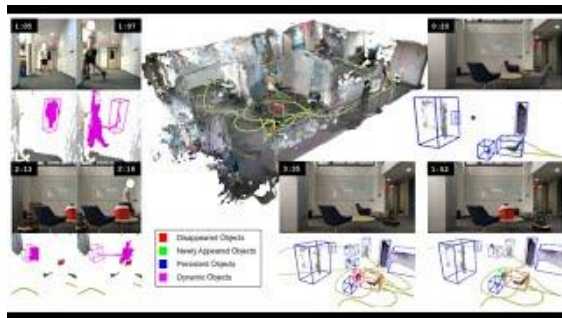
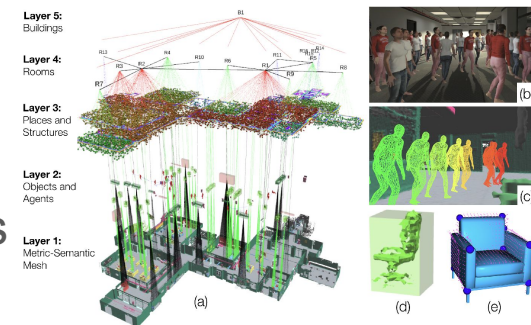
- ❖ Point Clouds
- ❖ Polygons / patches
- ❖ **Occupancy grid / Voxels**
- ❖ Semantics / objects
- ❖ NeRFs
- ❖ Gaussian Splatting
- ❖ Foundation Models
- ❖ ...



What is SLAM? - Mapping

Representations:

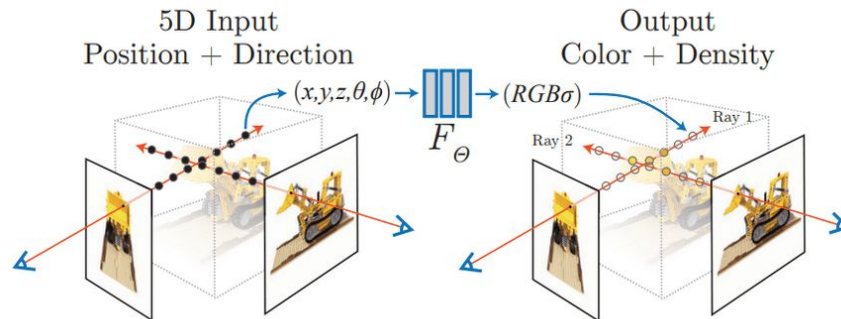
- ❖ Point Clouds
- ❖ Polygons / patches
- ❖ Occupancy grid / Voxels
- ❖ **Semantics / objects**
- ❖ NeRFs
- ❖ Gaussian Splatting
- ❖ Foundation Models
- ❖ ...



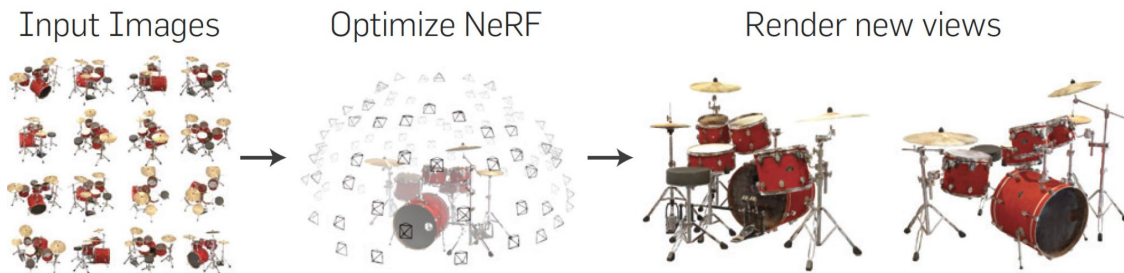
What is SLAM? - Mapping

Representations:

- ❖ Point Clouds
- ❖ Polygons / patches
- ❖ Occupancy grid / Voxels
- ❖ Semantics / objects
- ❖ **NeRFs**
- ❖ Gaussian Splatting
- ❖ Foundation Models
- ❖ ...



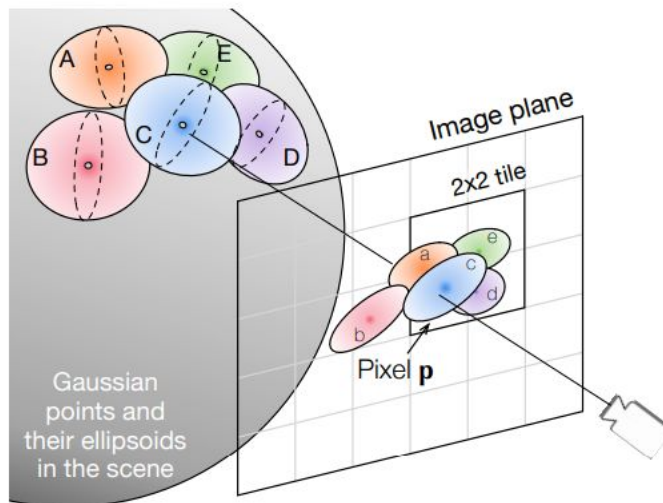
Neural Radiance Fields. Source: Mildenhall et al. ²



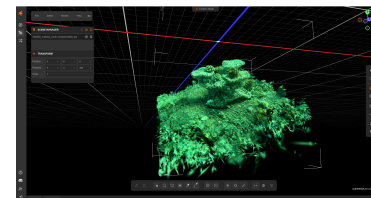
What is SLAM? - Mapping

Representations:

- ❖ Point Clouds
- ❖ Polygons / patches
- ❖ Occupancy grid / Voxels
- ❖ Semantics / objects
- ❖ NeRFs
- ❖ **Gaussian Splatting**
- ❖ Foundation Models
- ❖ ...



<https://huggingface.co/spaces/Bratanov/PineCreek-TheRock>
<https://superspl.at/editor>

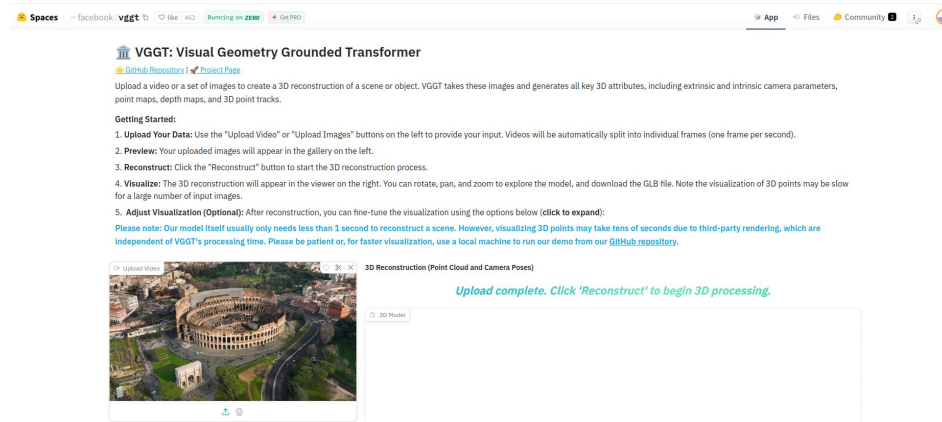


What is SLAM? - Mapping

Representations:

- ❖ Point Clouds
- ❖ Polygons / patches
- ❖ Occupancy grid / Voxels
- ❖ Semantics / objects
- ❖ NeRFs
- ❖ Gaussian Splatting
- ❖ **Foundation Models**
- ❖ ...

<https://huggingface.co/spaces/facebook/vggt>



[DUST3R: Geometric 3D Vision Made Easy](#)
[Grounding Image Matching in 3D with MAST3R](#)
[VGGT: Visual Geometry Grounded Transformer](#)



What is SLAM? - Localization

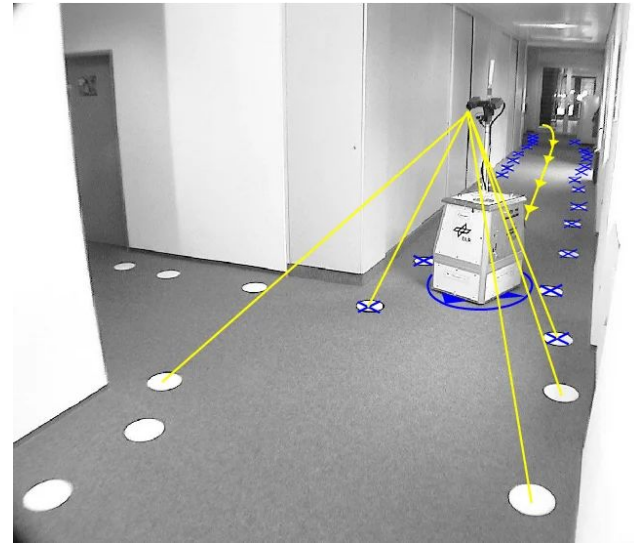
Localization is the process of determining precise position & orientation (aka **pose**) within the environment.

Localization can be:

- In a generated map
- In a given map—**kidnapped robot problem**

Localization Techniques:

- ❖ Kalman Filters
- ❖ Particle Filters
- ❖ Graph Optimization



Simultaneous Localization and Mapping

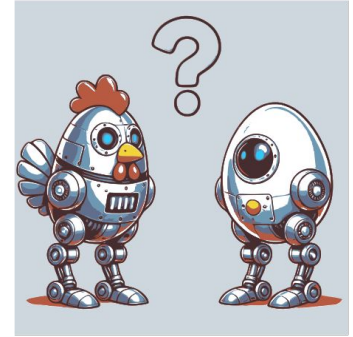
Simultaneous Localization and Mapping —better known as SLAM— refers to the fundamental problem of building spatial models of an environment while simultaneously determining the position of a robot within that environment.

- SLAM reduces to localization if the map is given, in which case the robot only has to estimate its pose with respect to the map.
- SLAM reduces to mapping if the pose of the robot is already known, for instance when an absolute positioning system is used (e.g., differential GPS or motion capture), in which case the robot only needs to model its surroundings using its sensor data.

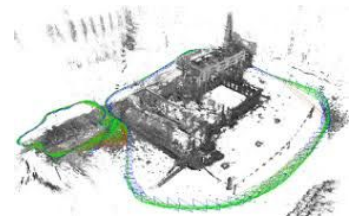


Anatomy of a Modern SLAM System

The ultimate goal of SLAM is to infer a map representation and robot poses (i.e., the robot trajectory) from sensor data, including data from proprioceptive sensors (e.g., wheel odometry or inertial measurement unit, IMU) and exteroceptive sensors (e.g., cameras, light detection and ranging (LiDAR), radars).



In mathematical terms this can be understood as an inverse problem: given a set of measurements, the goal is to determine a model of the world (the **map**) and a set of robot poses (**trajectory**) that could have produced those measurements.



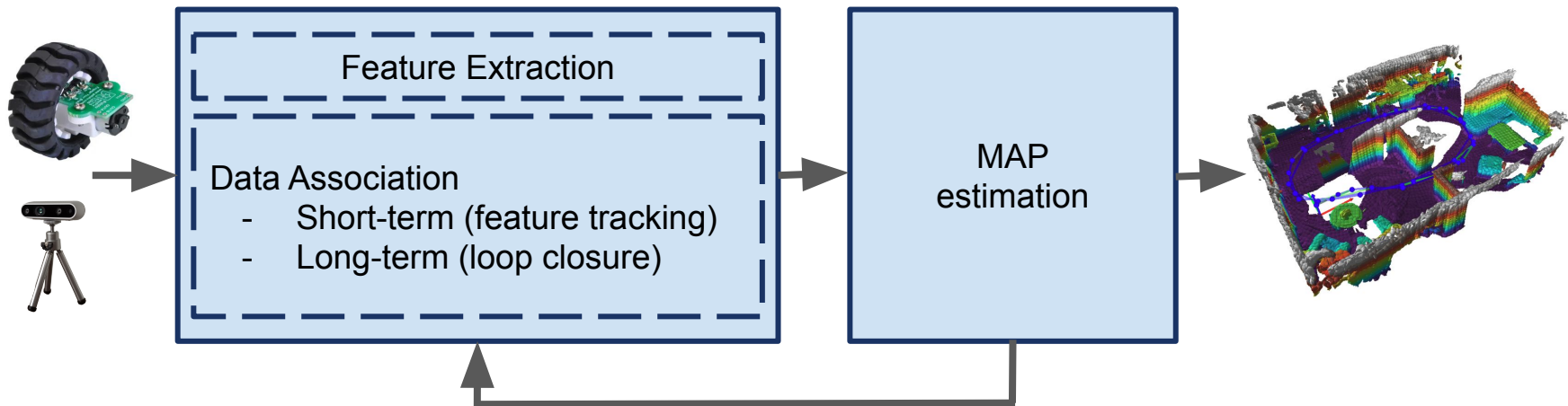
Anatomy of a Modern SLAM System

Sensors

Front-End

Back-End

SLAM Estimate



- Front-end algorithms: real-time and online processing of sensor data, trajectory, map
- Back-end algorithms: optimization and refinement of map and trajectory



Anatomy of a Modern SLAM System - Sensors

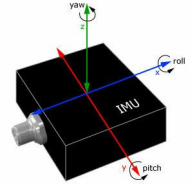
Proprioceptive sensors measure internal state of the robot providing information about position, orientation, and movement.

Exteroceptive sensors gather information about robot's environment detecting external objects, obstacles, and conditions.

Wheel Encoder



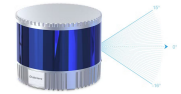
IMU



GPS / GNSS



LiDAR



Cameras

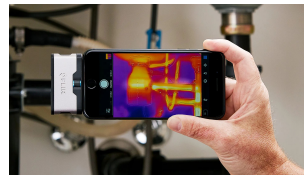
RGB



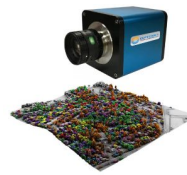
Stereo



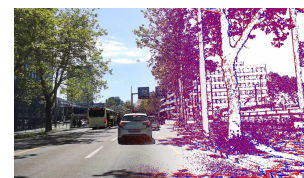
Thermal



Hyperspectral



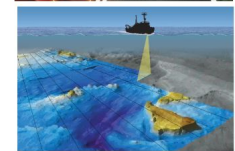
Event



Radar



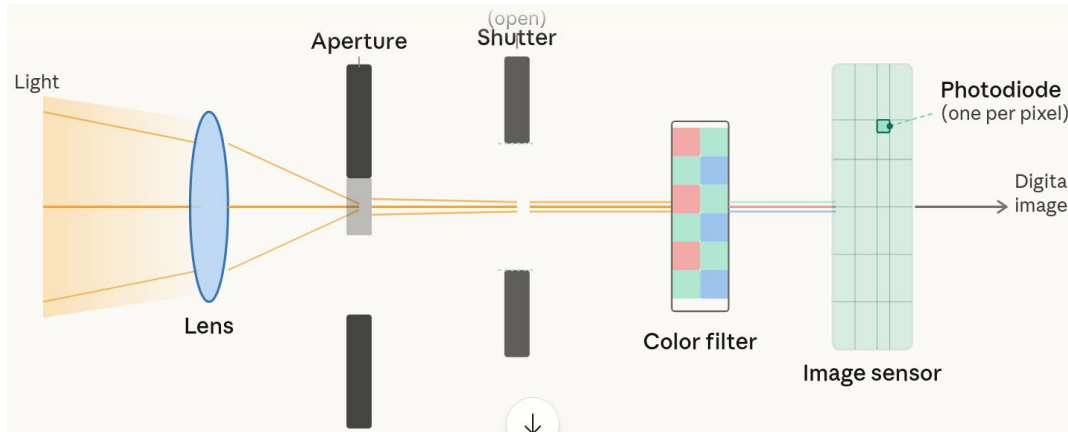
Sonar



Anatomy of a Modern SLAM System - Cameras

Cameras are the most commonly used sensors in SLAM/robotics. Serve as eyes of robots.

Monocular (mono) or RGB cameras capture red, green, and blue spectrums of the visible light



Anatomy of a Modern SLAM System - Cameras

Advantages:

- Cost-effective, affordable, widely available
- (vs other sensors)
- Rich visual data provide detailed info of the environment
- Small size/weight

Challenges:

- Poor performance in low-light conditions
- Motion blur with fast motion
- High computational need to process image data
- Projective/2D geometry; scale ambiguity



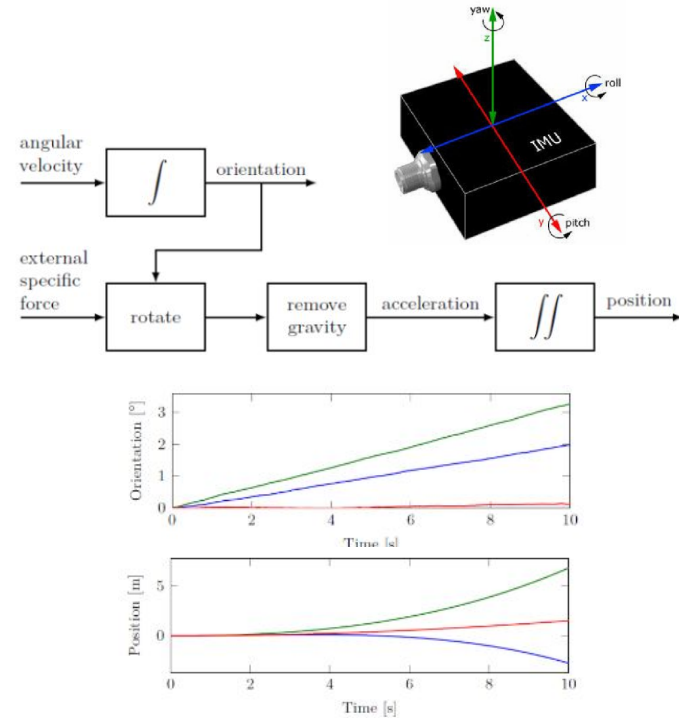
Anatomy of a Modern SLAM System - IMU

Inertial Measurement Unit (IMU)

Components:

- **Accelerometer**: linear acceleration
- **Gyroscope**: angular velocity
- Magnetometer: strength and direction of the magnetic field

The process of integrating the IMU measurements to get position and orientation estimates is called **dead-reckoning**.



Anatomy of a Modern SLAM System - IMU

IMU grades:

- **Consumer:** cell phones < \$100
- **Industrial:** vehicles and robotics \$100 - \$1000
- **Tactical:** military and robotics \$5000 - \$25000
- **Navigational:** commercial airlines, satellites, space > \$100000

Rough idea for accelerometer error:

$$e_x = \frac{1}{2} \times b \times t^2$$

- At 100 mg, loses its ability to give 50-meter accuracy after around ? seconds
- At 10 ug, loses its ability to give 50-meter accuracy after around ? minutes



Anatomy of a Modern SLAM System - IMU

IMU grades:

- **Consumer:** cell phones < \$100
- **Industrial:** vehicles and robotics \$100 - \$1000
- **Tactical:** military and robotics \$5000 - \$25000
- **Navigational:** commercial airlines, satellites, space > \$100000

Rough idea for accelerometer error:

$$e_x = \frac{1}{2} \times b \times t^2$$

- At 100 mg, loses its ability to give 50-meter accuracy after around **10 seconds**
- At 10 ug, loses its ability to give 50-meter accuracy after around **17 minutes**



Anatomy of a Modern SLAM System - IMU

Advantages:

- ❖ Proprioceptive (work in any environment)
- ❖ Small size/weight
- ❖ High measurement rate (1000Hz common) can capture fast motion

Challenges:

- ❖ Motion estimates drift due to (unobservable) biases and noise
- ❖ Sensitive to vibration, temperature change, magnetic interference
- ❖ High quality IMUs are expensive

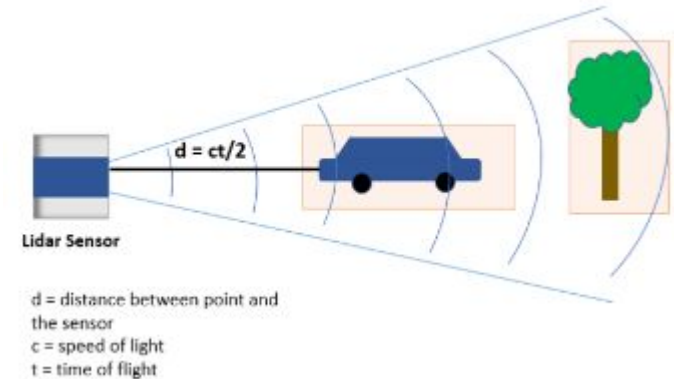


Anatomy of a Modern SLAM System - LiDAR

Light Detection and Ranging (LiDAR)

LiDAR calculates how long it takes for beams of light to hit an object and reflect back to the sensor -> distance to object

Common LiDAR sensors provide 2D or 3D scans
LiDARs can be rotating vs. solid state



Anatomy of a Modern SLAM System - LiDAR

Advantages:

- ❖ Works day/night
- ❖ Very high accuracy
- ❖ Provides 3D/depth information;
- ❖ Operates effectively over long distances



Challenges:

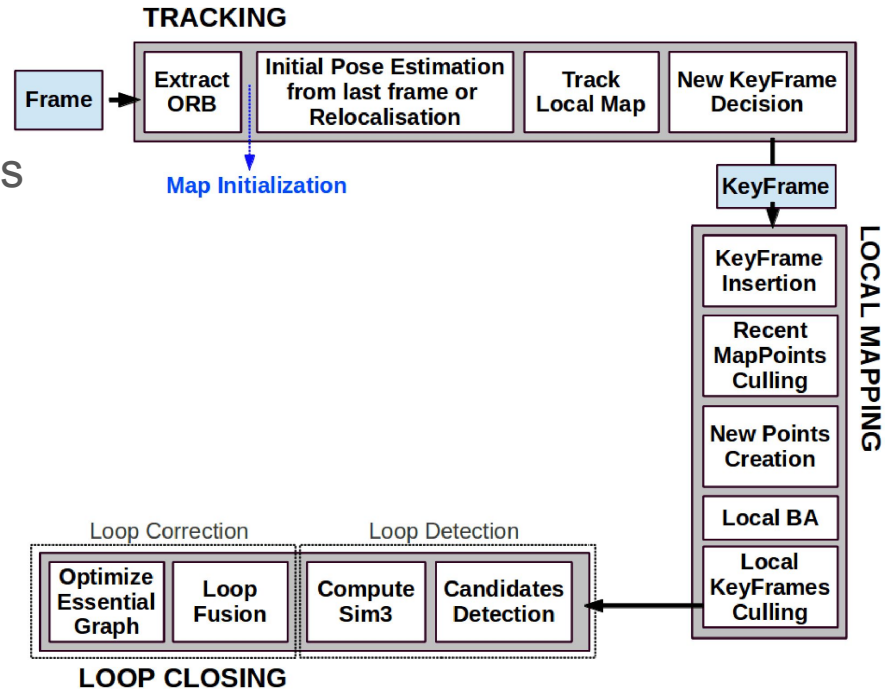
- ❖ Poor performance in rain, snow, fog, dust
- ❖ Surfaces with low reflectivity (black) may be challenging to detect
- ❖ Heavy / high power consumption
- ❖ More expensive than alternative sensors
- ❖ Crosstalk from other LiDAR devices



Anatomy of a Modern SLAM System - Front-End

Front-End

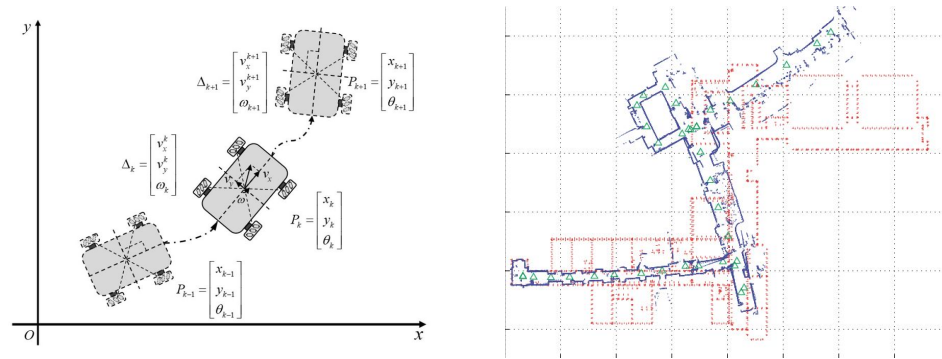
- Process of sensor data
- Extracts intermediate representations
- Data association
- Builds an initial guess
 - Trajectory / pose estimate (**odometry**)
 - Initial map estimate
- Deals with **real-time** & **online**



Anatomy of a Modern SLAM System - Odometry

Odometry is the use of data from motion sensors to estimate a moving robot's change in position over time.

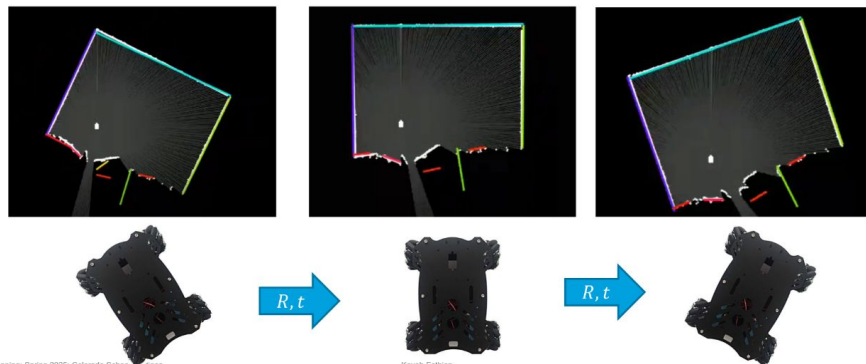
The key limitation is **drift**: small errors in each measurement accumulate over time, so the estimated position gradually diverges from the true position.



Anatomy of a Modern SLAM System - LiDAR Odometry

LiDAR Odometry

- Objective: Estimate robot trajectory/pose from **consecutive LiDAR scans** and create a primitive 3D map
- Associate corresponding points across scans – **how?** Estimate the rigid transformation (R, t) from correspondences



Robotic Mapping, Spring 2025, Colorado School of Mines

Kaveh Fathian

Arun's method provides least-square solution [Arun, PAMI87]

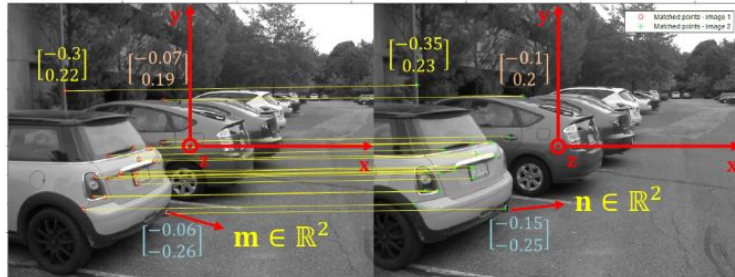
$$\min_{R,t} \sum_{i=1}^N ||p'_i - Rp_i + t||^2$$



Anatomy of a Modern SLAM System - Visual Odometry

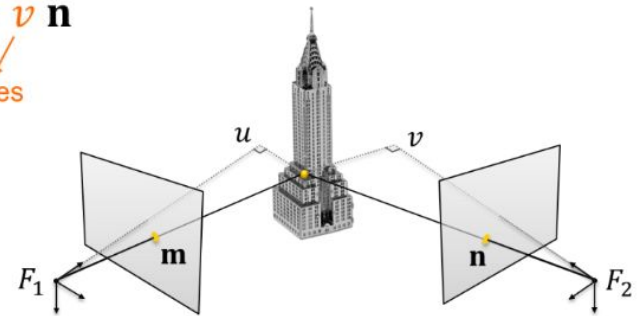
Visual Odometry

- Objective: Estimate robot **relative pose** from **consecutive images** and **reconstruct/triangulate features in 3D** – how?
- Associate 2D features across images (data association)
- Relative pose estimation: based on “rigid motion constraint” for each pair of matched feature keypoints:



$$u \mathbf{R} \mathbf{m} + \mathbf{t} = v \mathbf{n}$$

Unknown variables



SLAM in History

Erwin Wilhelm Kruppa (1885 - 1967)

Kruppa 1913:

Two views of five points are sufficient to determine both the camera motion and the 3D location of these points up to finitely many solutions.



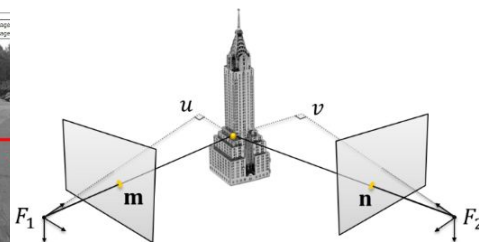
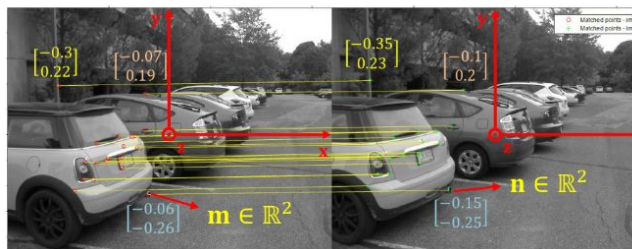
Anatomy of a Modern SLAM System - Visual Odometry

Visual Odometry

- Objective: Estimate robot **relative pose** from **consecutive images** and **reconstruct/triangulate features in 3D** – how?
- Associate 2D features across images (data association)
- Relative pose estimation: based on “rigid motion constraint” for each pair of matched feature keypoints:

Algorithms:

- 8pt algorithm [Longuet-Higgins, Nature81]
- 4pt algorithm (homography) [Hartley, ECCV94]
- 5pt algorithms [Nister, CVPR03; Stewenius JPRS06; Li ICPR06: Kukelova BMVC08]
- Direct estimation [Kneip Springer12]
- QuEst [Fathian RAL18]



Anatomy of a Modern SLAM System - Odometry

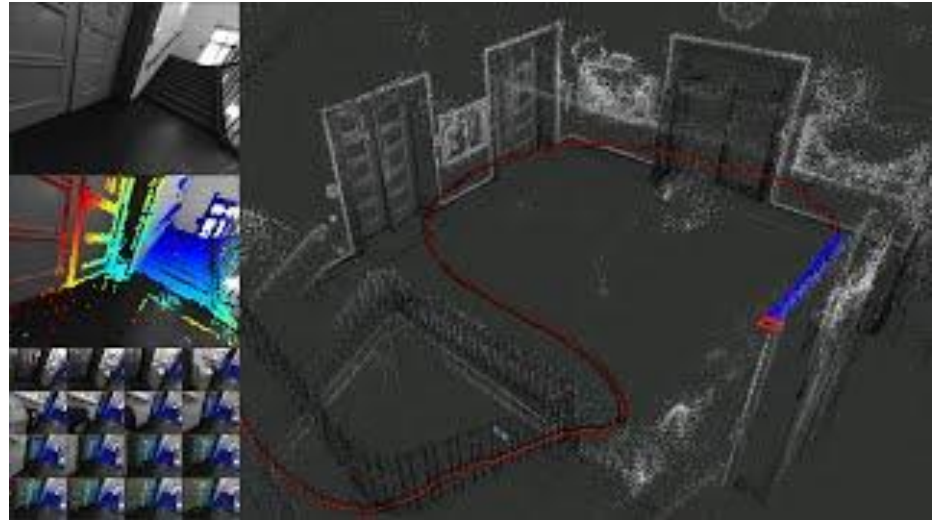
Direct Methods

- superior accuracy
- create additional non-convexities
- computationally cumbersome
- Photometric Loss

$$E_{\mathbf{p}j} := \sum_{\mathbf{p} \in \mathcal{N}_{\mathbf{p}}} w_{\mathbf{p}} \left\| (I_j[\mathbf{p}'] - b_j) - \frac{t_j e^{a_j}}{t_i e^{a_i}} (I_i[\mathbf{p}] - b_i) \right\|_{\gamma}$$

Indirect Methods

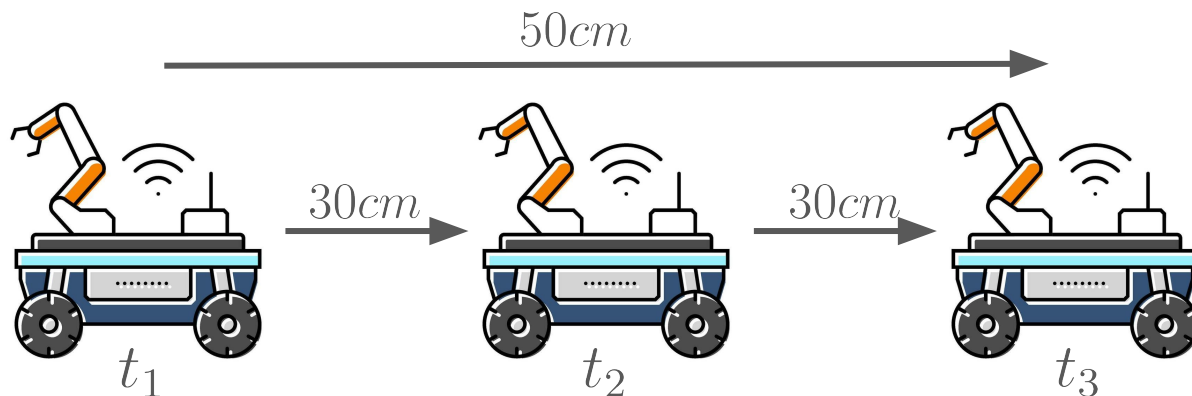
- faster
- memory efficient



Anatomy of a Modern SLAM System - Back-End

Geometric Consistency Example: Suppose we compute odometry estimates between (t_1, t_2) (t_2, t_3) (t_1, t_3)

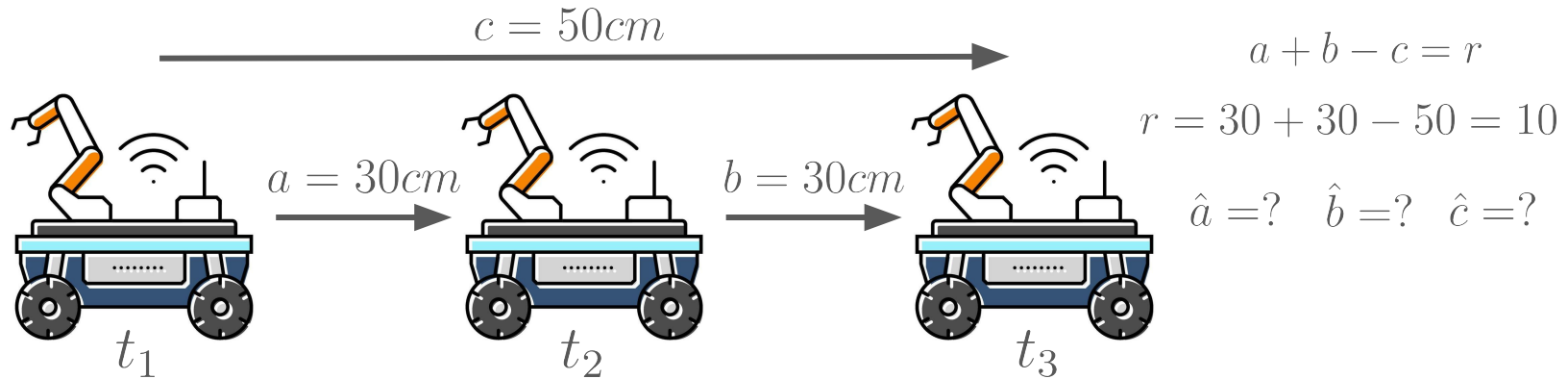
Estimates can be geometrically inconsistent (always are in practice!)



Anatomy of a Modern SLAM System - Back-End

Geometric Consistency Example: Suppose we compute odometry estimates between (t_1, t_2) (t_2, t_3) (t_1, t_3)

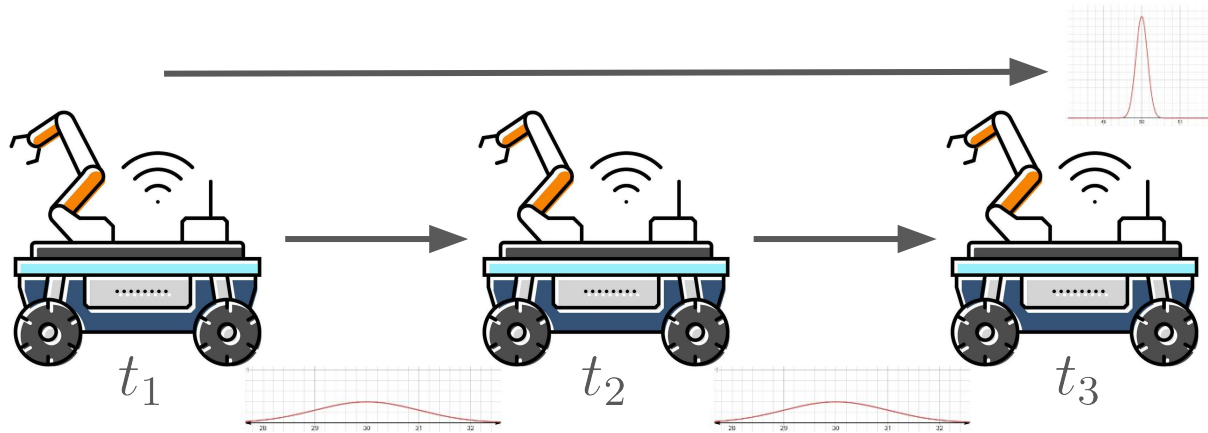
We can take average of the estimates to obtain consistent results



Anatomy of a Modern SLAM System - Back-End

Geometric Consistency Example: Suppose we compute odometry estimates between (t_1, t_2) (t_2, t_3) (t_1, t_3)

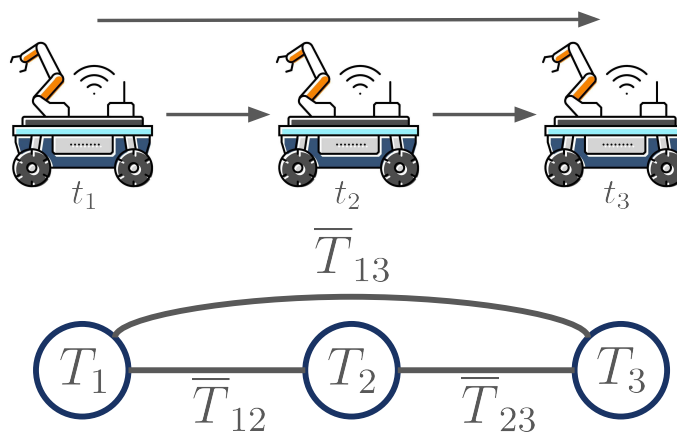
We can take average of the estimates to obtain consistent results



Anatomy of a Modern SLAM System - Back-End

Geometric Consistency Example: Suppose we compute odometry estimates between $(t_1, t_2)(t_2, t_3)(t_1, t_3)$

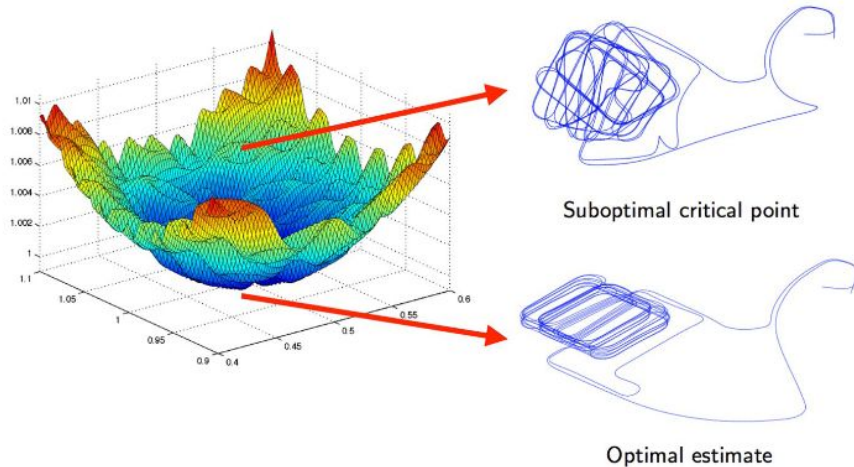
Factor Graph Formulation



Anatomy of a Modern SLAM System - Back-End

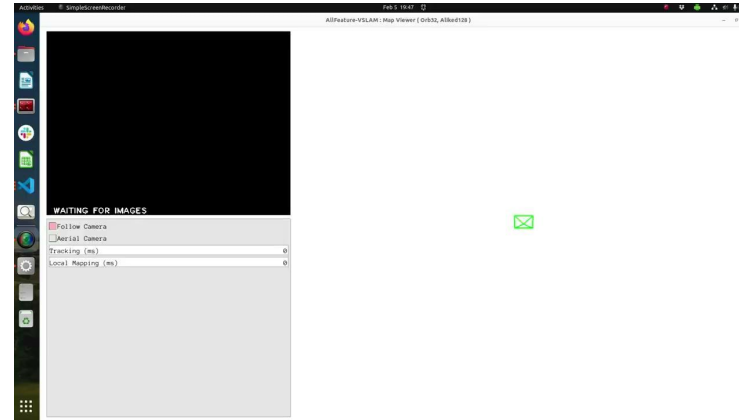
Back-end deals with:

- Optimization/refinement of map and trajectory (for consistency)
- Finding loop closures correcting drift



Anatomy of a Modern SLAM System - Loop Closure

Loop Closure



- ❖ Image Retrieval / Visual Place Recognition (VPR)
- ❖ Impose new constraints (Pose Graph)
- ❖ Global Optimization (Global Bundle Adjustment)



Anatomy of a Modern SLAM System - Loop Closure

Image Retrieval / Visual Place Recognition (VPR)

Approaches:

Local Descriptors

- Allow estimating feature and camera geometry
- Sensitive to lighting and seasonal variations

Global Descriptors

- Better at handling lighting and seasonal variations
- More sensitive to viewpoint changes

Learning-based methods



(a)

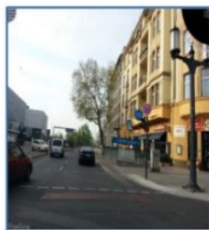


Anatomy of a Modern SLAM System - Loop Closure

Image Retrieval / Visual Place Recognition (VPR)

Challenges:

- Perceptual Aliasing
- Appearance Changes
- Dynamic Objects



Viewpoint Variation



Perceptual Aliasing



Seasonal Change



Day/Night Change



Weather Change

Wrong loop closure is catastrophic! -> need outlier removal



Anatomy of a Modern SLAM System - Ex. ORB-SLAM2

Sensors:

Front-End:

Back-End:

SLAM Estimate:



Anatomy of a Modern SLAM System - Ex. ORB-SLAM2

Sensors:

- Monocular Camera

Front-End:

- Feature-based
- Loop closure

Back-End:

- Factor-Graph
- Non-linear optimization

SLAM Estimate:

- Point Cloud
- 6dof camera poses



Why Do We Need SLAM? Use GPS

- ❖ **Differential GPS and motion capture systems are expensive** and restricted to small areas, hence being unsuitable for large-scale robot deployments.
- ❖ **Consumer-grade GPS** is much more broadly available, but its accuracy (with errors typically in the order of meters) and its availability (which is limited to outdoor areas with line-of-sight to satellites) often makes it unsuitable as a source of localization;



Why do we need SLAM? We need a map

- ❖ In most robotics applications, **prior maps are unavailable**, making SLAM necessary rather than pure localization.
- ❖ In some cases, **map-building is the primary mission goal** — such as disaster response robots charting unknown environments to aid first responders.
- ❖ Even when **partial maps** exist, they are often **too outdated** or coarse to be useful — a home floor plan won't capture furniture arrangements, and Mars rovers' satellite maps **lack the resolution** needed for local obstacle avoidance and motion planning.



Do We Really Need SLAM for Robotics?

Do we need SLAM for any robotics task? No

- ❖ Reactive tasks: keeping a target in sight, can be solved with simpler control strategies (e.g. visual servoing)
- ❖ A robot that has to operate over small distances, relying on odometry estimates and local mapping might be acceptable
- ❖ The environment the robot operates in has some infrastructure for localization.

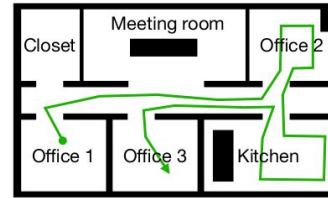
SLAM seems an indispensable component for long-term operation in unstructured environments: **long term operation** typically requires **memory** (e.g., to go back to previously seen objects or find suitable collision-free paths), and map representations



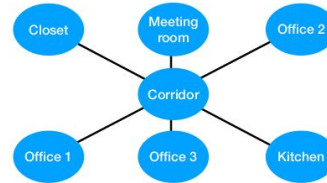
Do We Really Need SLAM for Robotics?

Do we need globally consistent geometric maps for navigation?

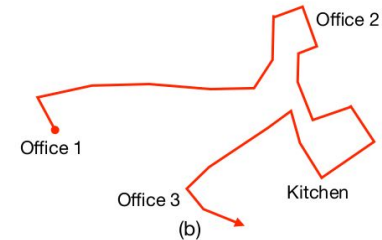
Imagine that our robot visits Office 1 in a building and then, after exploring other areas of the building, it visits Office 3, which is just next door from Office 1. Using just odometry, the robot might be misled to conclude that Office 1 and Office 3 are quite far from each other (due to the odometry drift), hence being unable to realize there is a short path connecting the two offices



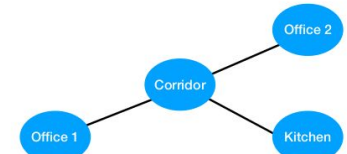
(a)



(c)



(b)



(d)



Do We Really Need SLAM for Robotics?

Do we need maps?

One might take a different approach: in order for the robot to execute a task, the robot might be trained to translate raw sensor data directly to actions (e.g., using **Reinforcement Learning**), hence circumventing the need to build a map.

In such an approach, the neural network trained from sensor data to actions will arguably create an internal representation, but such an internal representation cannot be directly queried, inspected, or visualized.

While the jury is still out on whether maps are indeed necessary, there is some initial evidence that using maps as an **intermediate representation** is at least beneficial in completing many visual tasks for robotics.



SLAM Applications



SLAM Applications



Autonomous Driving



Augmented Reality



Virtual Reality



Search & Rescue



Planetary Exploration



Humanoids



Healthcare



Industrial Robots



Vacuum Cleaners



Coral Monitoring



Mining



Delivery



Drone Racing



Agriculture



Disability Support



Defence



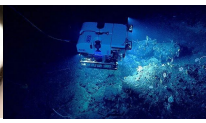
Construction



Warehouses



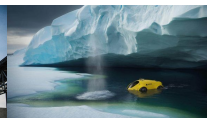
Pipe Inspection



Map Ocean Floor



Inspection



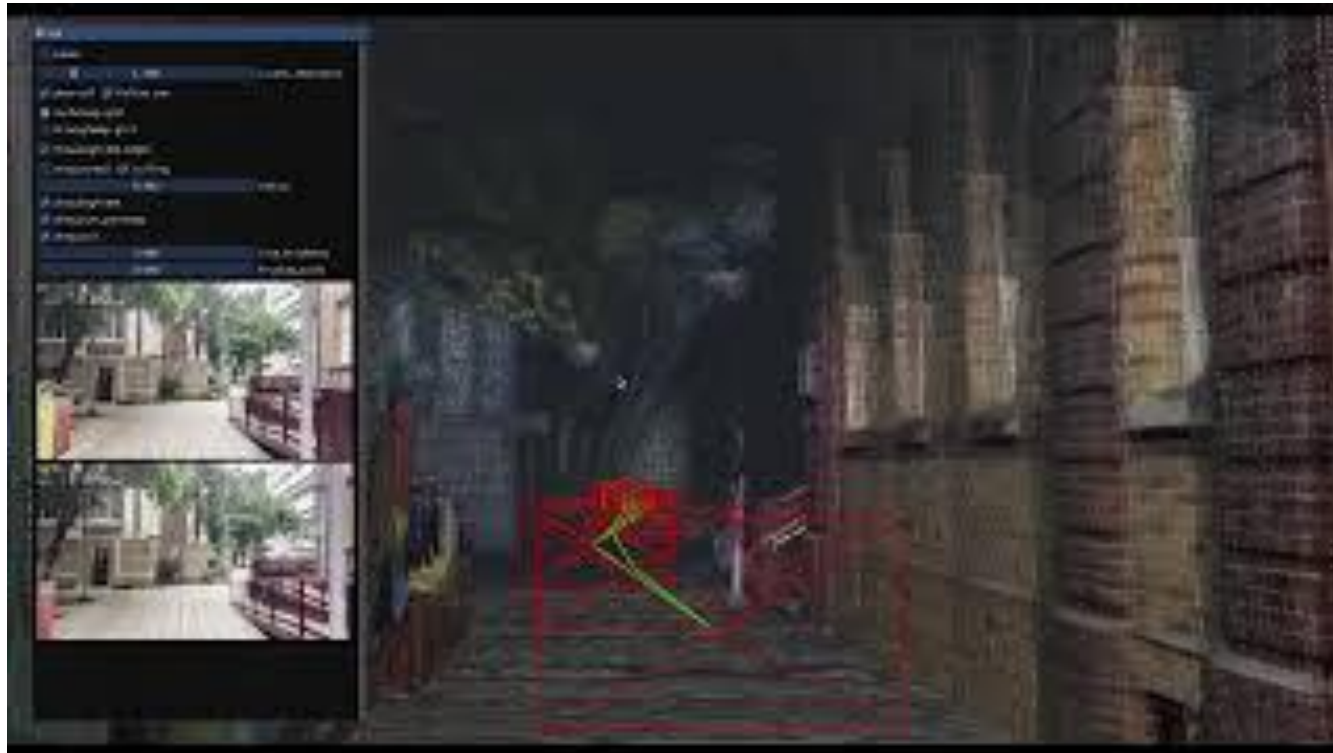
Glacier Tracking



Filmmaking



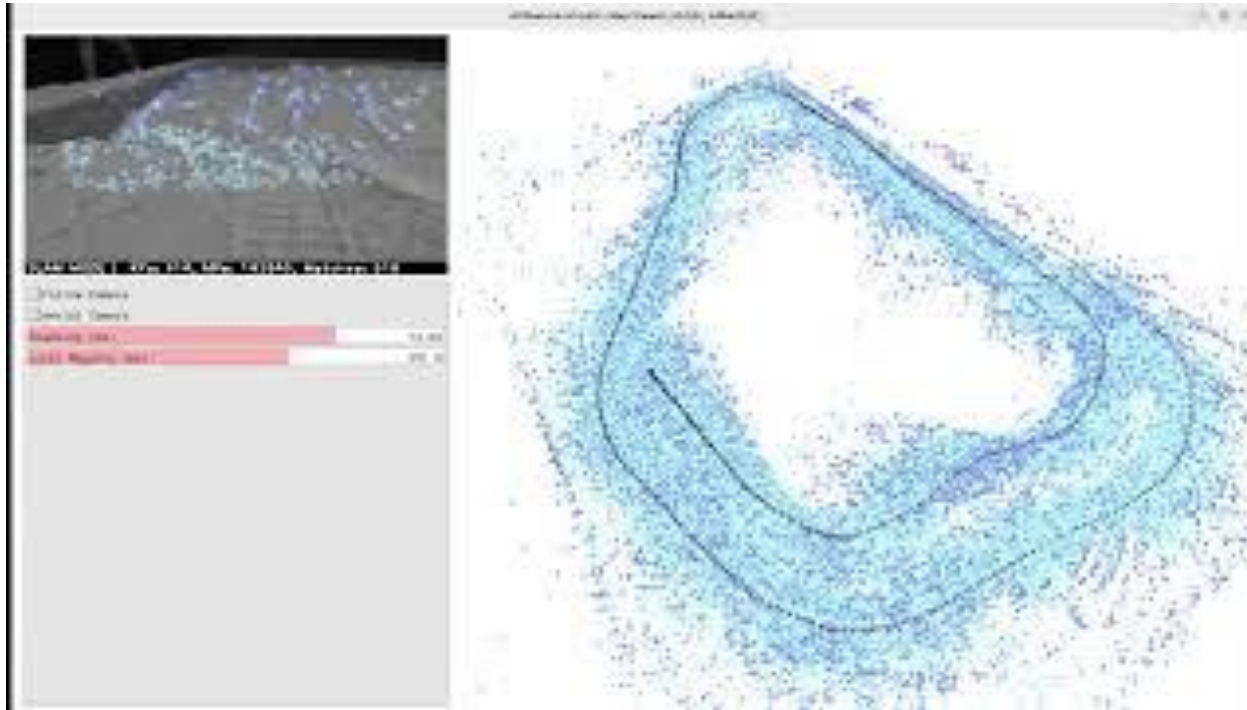
SLAM Examples - MAST3R-SLAM



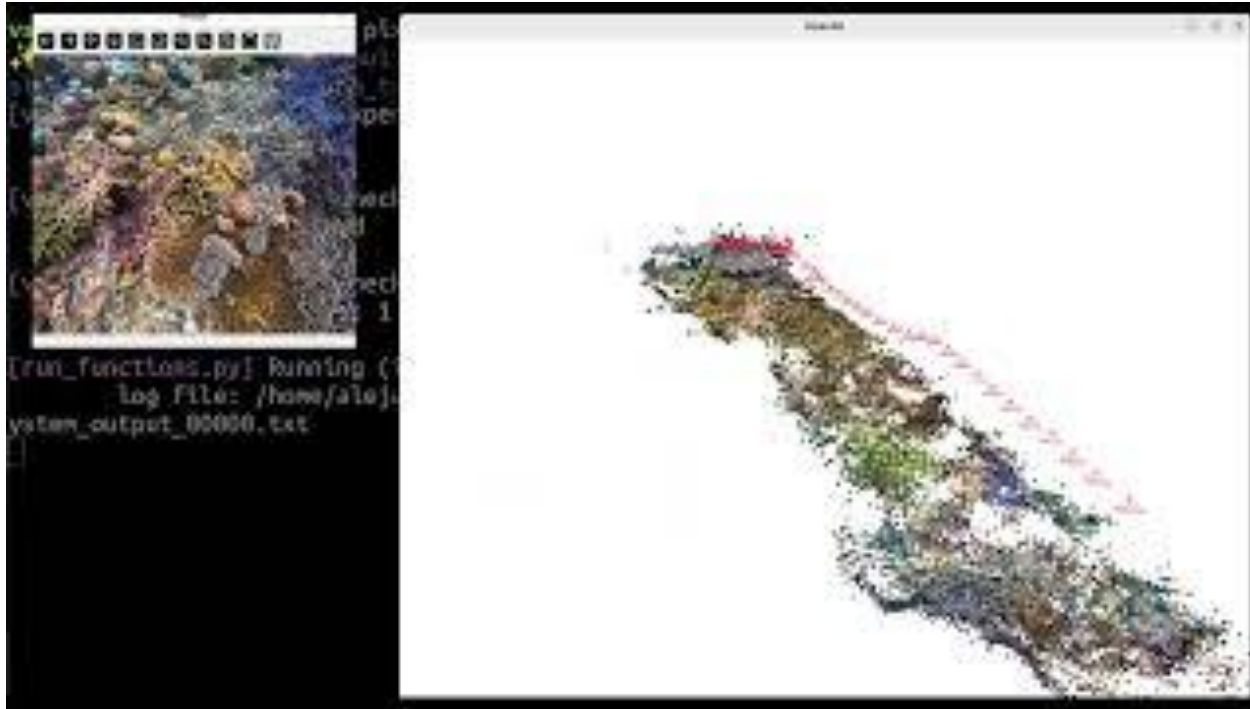
SLAM Examples - Sensor Modalities



SLAM Examples - All-Feature-VSLAM



SLAM Examples - All-Feature-VSLAM



Simultaneous Localization and Mapping



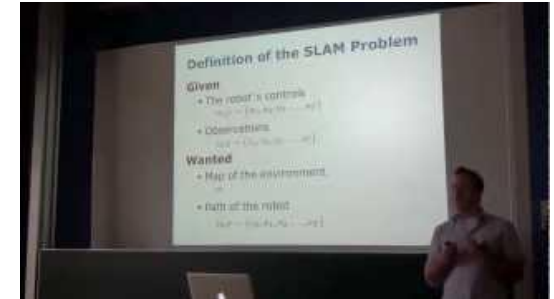
Localization- 5 Minutes with Cyrill

<https://youtu.be/TLe4otkRAts?list=PLgnQpQtFTOGSO8HC48K9sPuNliY1qxzV9>



SLAM- 5 Minutes with Cyrill

<https://youtu.be/BuRCJ2fegcc?list=PLgnQpQtFTOGSO8HC48K9sPuNliY1qxzV9>



SLAM Course (Cyrill Stachniss)

<https://youtu.be/U6vr3iNrwRA?list=PLgnQpQtFTOGQrZ4O5QzblHgl3b1JHimN>



SLAM Handbook

<https://github.com/SLAM-Handbook-contributors/slam-handbook-public-release>

Probabilistic Robotics

https://www.academia.edu/6819774/PROBABILISTIC_ROBOTICS



Robotic Mapping and Localization

<https://github.com/ariarobotics/robotic-mapping>

VSLAM-LAB

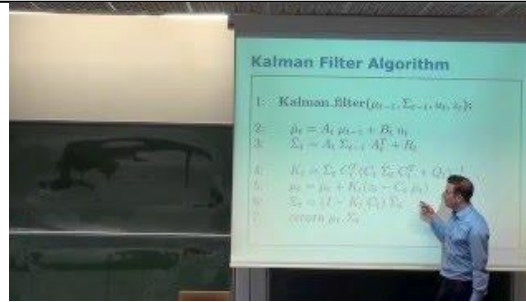
<https://github.com/VSLAM-LAB/VSLAM-LAB>



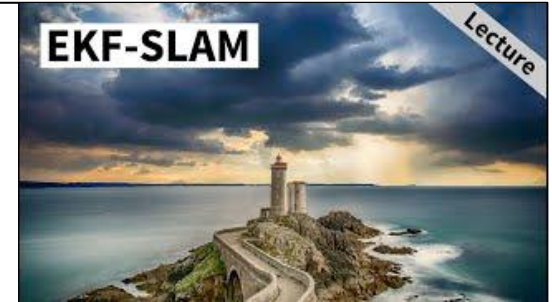
Kalman Filter, EKF & EKF-SLAM - Tomorrow



Kalman Filter - 5 Minutes with Cyrill
https://youtu.be/o_HW6GnLqvq



Photogrammetry II - 12 - Kalman Filter and EKF (2015/16)
https://youtu.be/BblUVCcT__s



EKF-SLAM (Cyrill Stachniss)
<https://youtu.be/X30sEglws0g>



SLAM Handbook
<https://github.com/SLAM-Handbook-contributors/slam-handbook-public-release>

Probabilistic Robotics
https://www.academia.edu/6819774/PROBABILISTIC_ROBOTICS

